

TRIGONOMETRY

SINGLE CORRECT TYPE QUESTIONS

1. The value of $\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$ is :

$\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$ का मान है :

- (A) $\frac{3}{4}$ (B) $\frac{3}{2}(1 + \cos 20^\circ)$ (C) $\frac{3}{2}$ (D) $\frac{3}{4} + \cos 20^\circ$

Ans. A

sol.
$$\begin{aligned} & \left(\frac{1 + \cos 20^\circ}{2} \right) + \left(\frac{1 + \cos 100^\circ}{2} \right) - \frac{1}{2} (2 \cos 10^\circ \cos 50^\circ) \\ &= 1 + \frac{1}{2} (\cos 20^\circ + \cos 100^\circ) - \frac{1}{2} [\cos 60^\circ + \cos 40^\circ] \\ &= \left(1 - \frac{1}{4} \right) + \frac{1}{2} [\cos 20^\circ + \cos 100^\circ - \cos 40^\circ] \\ &= \frac{3}{4} + \frac{1}{2} [2 \cos 60^\circ \times \cos 40^\circ - \cos 40^\circ] \\ &= \frac{3}{4} \end{aligned}$$

2. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in :

$\theta \in [0, 2\pi]$ के सभी संभव मान, जिनके लिए $\sin 2\theta + \tan 2\theta > 0$ है, निम्न में से किस में है ?

- (A) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ (B) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$
 (C) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$ (D) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

Ans. C

Sol. $\sin 2\theta + \tan 2\theta > 0$

$$\sin 2\theta + \frac{\sin 2\theta}{\cos 2\theta} > 0$$

$$\sin 2\theta \left(\frac{\cos 2\theta + 1}{\cos 2\theta} \right) > 0$$

$$\tan 2\theta(2\cos^2\theta) > 0$$

$$\Rightarrow \tan 2\theta > 0 \quad (\cos^2\theta \neq 0)$$

$$\theta \in [0, 2\pi]$$

$$2\theta \in [0, 4\pi]$$

$$2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(2\pi, \frac{5\pi}{2}\right) \cup \left(3\pi, \frac{7\pi}{2}\right)$$

$$\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$$

3. The value of $\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right)$ is :-

$$\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right) \text{ का मान है :}$$

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{1}{2\sqrt{2}}$

(D) $\frac{1}{\sqrt{2}}$

Ans. C

Sol. $\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right)$

$$\text{Let } \theta = \frac{\pi}{8}$$

$$= \cos^3\theta \cdot \cos(3\theta) + \sin^3\theta \cdot \sin(3\theta)$$

$$= \cos^3\theta(4\cos^3\theta - 3\cos\theta) + \sin^3\theta(3\sin\theta - 4\sin^3\theta)$$

$$= 4(\cos^6\theta - \sin^6\theta) + 3[\sin^4\theta - \cos^4\theta]$$

$$= 4(\cos^2\theta - \sin^2\theta)(\cos^4\theta + \cos^2\theta \cdot \sin^2\theta + \sin^4\theta) + 3(\sin^2\theta - \cos^2\theta) \cdot 1$$

$$= 4\cos 2\theta ((\cos^2\theta + \sin^2\theta)^2 - \sin^2\theta \cos^2\theta) - 3\cos 2\theta$$

$$= 4\cos 2\theta \left(1 - \frac{1}{4}\sin^2 2\theta\right) - \frac{3}{\sqrt{2}}$$

$$= \frac{4}{\sqrt{2}} \times \frac{7}{8} - \frac{3}{\sqrt{2}}$$

$$= \frac{7}{2\sqrt{2}} - \frac{6}{2\sqrt{2}} = +\frac{1}{2\sqrt{2}}$$

4. For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ the expression $3(\sin \theta - \cos \theta)^4 + 6(\sin \theta + \cos \theta)^2 + 4\sin^6 \theta$ equals :

किसी $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ के लिए व्यंजक $3(\sin \theta - \cos \theta)^4 + 6(\sin \theta + \cos \theta)^2 + 4\sin^6 \theta$ बराबर है :

- (A) $13 - 4 \cos^2 \theta + 6 \sin^2 \cos^2 \theta$ (B) $13 - 4 \cos^4 \theta + 2 \sin^2 \cos^2 \theta$
(C) $13 - 4 \cos^2 \theta + 6 \cos^4 \theta$ (D) $13 - 4 \cos^6 \theta$

Ans. D

Sol. $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$$\begin{aligned} & 3(1 - 2\sin \theta \cos \theta)^2 + 6(1 + 2\sin \theta \cos \theta) + 4 \sin^6 \theta \\ \Rightarrow & 3(1 + 4\sin^2 \theta \cos^2 \theta - 4 \sin \theta \cos \theta) + 6 + 12\sin \theta \cos \theta + 4\sin^6 \theta \\ \Rightarrow & 3 + 12\sin^2 \theta \cos^2 \theta + 6 + 4\sin^6 \theta \\ \Rightarrow & 9 + 12 \cos^2 \theta (1 - \cos^2 \theta) + 4(1 - \cos^2 \theta)^3 \\ \Rightarrow & 9 + 12 \cos^2 \theta - 12 \cos^4 \theta + 4(1 - \cos^6 \theta + 3\cos^4 \theta - 3\cos^2 \theta) \\ \Rightarrow & 13 - 4\cos^6 \theta \end{aligned}$$

5. If $L = \sin^2 \left(\frac{\pi}{16}\right) - \sin^2 \left(\frac{\pi}{8}\right)$ and $M = \cos^2 \left(\frac{\pi}{16}\right) - \sin^2 \left(\frac{\pi}{8}\right)$, then :

यदि $L = \sin^2 \left(\frac{\pi}{16}\right) - \sin^2 \left(\frac{\pi}{8}\right)$ तथा $M = \cos^2 \left(\frac{\pi}{16}\right) - \sin^2 \left(\frac{\pi}{8}\right)$ हैं, तो :

- (A) $L = \frac{-1}{2\sqrt{2}} + \frac{1}{2} \cos \frac{\pi}{8}$ (B) $M = \frac{1}{2\sqrt{2}} + \frac{1}{2} \cos \frac{\pi}{8}$
(C) $M = \frac{1}{4\sqrt{2}} + \frac{1}{4} \cos \frac{\pi}{8}$ (D) $L = \frac{1}{4\sqrt{2}} - \frac{1}{4} \cos \frac{\pi}{8}$

Ans. B

Sol.
$$L = \left(\frac{1 - \cos \frac{\pi}{8}}{2} \right) - \left(\frac{1 - \cos \frac{\pi}{4}}{2} \right)$$

$$L = \frac{1}{2\sqrt{2}} - \frac{1}{2} \cos \frac{\pi}{8}$$

$$M = \left(\frac{1 + \cos \frac{\pi}{8}}{2} \right) - \left(\frac{1 - \cos \frac{\pi}{4}}{2} \right)$$

$$= \frac{1}{2\sqrt{2}} + \frac{1}{2} \cos \frac{\pi}{8}$$

6. The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is :

$36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ का मान है :

- (A) 54 (B) 27 (C) 18 (D) 36

Ans. D

Sol. $4\cos^2 \theta - 1 = 4(1 - \sin^2 \theta) - 1 = 3 - 4\sin^2 \theta = \frac{\sin 3\theta}{\sin \theta}$

so given expression can be written as

$$36 \times \frac{\sin 27^\circ}{\sin 9^\circ} \times \frac{\sin 81^\circ}{\sin 27^\circ} \times \frac{\sin 243^\circ}{\sin 81^\circ} \times \frac{\sin 729^\circ}{\sin 243^\circ}$$

$$36 \times \frac{\sin 729^\circ}{\sin 9^\circ} = 36$$

7. The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x , is

λ के सभी मानों, जिनके लिए समीकरण $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ का एक वास्तविक हल x है, का समुच्चय है :

- (A) $\left[-1, -\frac{1}{2}\right]$ (B) $\left[-2, -\frac{3}{2}\right]$ (C) $[-2, -1]$ (D) $\left[-\frac{3}{2}, -1\right]$

Ans. D

Sol. $\lambda = \cos^2 2x - 2\sin^4 x - 2\cos^2 x$

convert all in to $\cos x$.

$$\lambda = (2\cos^2 x - 1)^2 - 2(1 - \cos^2 x)^2 - 2\cos^2 x$$

$$= 4\cos^4 x - 4\cos^2 x + 1 - 2(1 - 2\cos^2 x + \cos^4 x) - 2\cos^2 x$$

$$= 2\cos^4 x - 2\cos^2 x + 1 - 2$$

$$= 2\cos^4 x - 2\cos^2 x - 1$$

$$= 2 \left[\cos^4 x - \cos^2 x - \frac{1}{2} \right]$$

$$= 2 \left[\left(\cos^2 x - \frac{1}{2} \right)^2 - \frac{3}{4} \right]$$

$$\lambda_{\max} = 2 \left[\frac{1}{4} - \frac{3}{4} \right] = 2 \times \left(-\frac{2}{4} \right) = -1 \text{ (max Value)}$$

$$\lambda_{\min} = 2 \left[0 - \frac{3}{4} \right] = -\frac{3}{2} \text{ (minimum value)}$$

$$\text{So, Range} = \left[-\frac{3}{2}, -1 \right]$$

8. Let $S = \left\{ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10 \right\}$ and $\beta = \sum_{x \in S} \tan^2 \left(\frac{x}{3} \right)$, then $\frac{1}{6}(\beta - 14)^2$ is equal to :

माना $S = \left\{ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10 \right\}$ तथा $\beta = \sum_{x \in S} \tan^2 \left(\frac{x}{3} \right)$ हैं, तो $\frac{1}{6}(\beta - 14)^2$ बराबर है :

(A) 32

(B) 64

(C) 16

(D) 8

Ans. A

Sol. $9^{1-\tan^2 x} + 9^{\tan^2 x} = 10$

$$\Rightarrow \frac{9}{y} + y = 10 \text{ (where } 9^{\tan^2 x} = y)$$

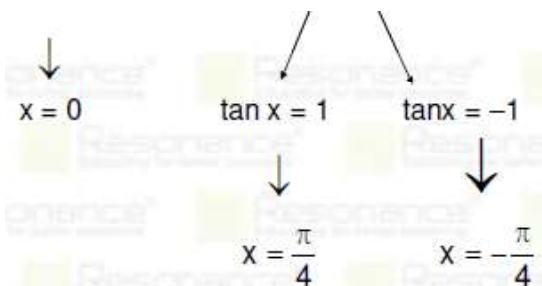
$$\text{So, } 9 + y^2 = 10y$$

$$y^2 - 10y + 9 = 0$$

$$(y-1)(y-9) = 0$$

$$\text{So, } 9^{\tan^2 x} = 1 \quad \& \quad 9^{\tan^2 x} = 9$$

$$\tan^2 x = 0 \quad \tan^2 x = 1$$



$$\text{Now } \beta = \sum \tan^2 \frac{x}{3} = \tan^2 \frac{0}{3} + \tan^2 \frac{\pi}{12} + \tan^2 \frac{\pi}{12}$$

$$= 2 \tan^2 \frac{\pi}{12}$$

$$= 2(2 - \sqrt{3})^2$$

$$= 2(7 - 4\sqrt{3})$$

$$\beta = 14 - 8\sqrt{3} \Rightarrow \frac{1}{6}(\beta - 14)^2 = 32$$

9. If $\sin^4 \alpha + 4\cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cos \beta$; $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to :

यदि $\sin^4 \alpha + 4\cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cos \beta$; $\alpha, \beta \in [0, \pi]$, तो $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ बराबर है -

- (A) 0 (B) $-\sqrt{2}$ (C) $\sqrt{2}$ (D) -1

Ans. B

sol. $\sin^4 \alpha + 4\cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cos \beta$; $\alpha, \beta \in [0, \pi]$

AM $\geq$ GM

$$\frac{\sin^4 \alpha + 4\cos^4 \beta + 1 + 1}{4} \geq \sqrt[4]{\sin^4 \alpha \cos^4 \beta \cdot 1 \cdot 1}$$

$$\sin^4 \alpha + 4\cos^4 \beta + 2 \geq 4\sqrt{2} \sin \alpha \cos \beta$$

AM = GM

So

$$\sin^4 \alpha = 4\cos^4 \beta = 1$$

$$\sin \alpha = 1 \quad \cos \beta = \frac{\pm 1}{\sqrt{2}}$$

$$\sin \beta = \frac{1}{\sqrt{2}} \quad \beta \in [0, \pi]$$

$$\cos(\alpha + \beta) - \cos(\alpha - \beta)$$

$$= -2 \sin \alpha \sin \beta$$

$$= -2 \frac{1}{\sqrt{2}}$$

$$-2\sqrt{2}$$

10. The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is :

समीकरण $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ के हलों की संख्या है :

- (A) 3 (B) 5 (C) 4 (D) 7

Ans. B

sol. $1 + \sin^4 x = \cos^2 3x$

$$\text{L.H.S} = 1 + \sin^4 x, \text{R.H.S} = \cos^2 3x$$

$$\text{L.H.S} \geq 1 \quad \text{R.H.S} \leq 1$$

$$\Rightarrow \text{L.H.S.} = \text{R.H.S.} = 1$$

$$\sin^4 x = 0 \text{ and } \cos^2 3x = 1$$

$$\sin x = 0 \text{ and } (4\cos^2 x - 3)^2 \cos^2 x = 1$$

$$\Rightarrow \sin x = 0 \text{ and } \cos^2 x = 1$$

$$\Rightarrow x = 0, \pm \pi, \pm 2\pi$$

$$\Rightarrow \text{Total number of solutions is 5}$$

11. The number of elements in the set $S = \{\theta \in [0, 2\pi] : 3\cos^4 \theta - 5\cos^2 \theta - 2\sin^6 \theta + 2 = 0\}$ is :

समुच्चय $S = \{\theta \in [0, 2\pi] : 3\cos^4 \theta - 5\cos^2 \theta - 2\sin^6 \theta + 2 = 0\}$ में अवयवों की संख्या है :

- (A) 9 (B) 10 (C) 8 (D) 12

Ans. A

Sol. $3\cos^4 \theta - 5\cos^2 \theta - 2\sin^6 \theta + 2 = 0$

$$\Rightarrow 3\cos^4 \theta - 3\cos^2 \theta - 2\cos^2 \theta - 2\sin^6 \theta + 2 = 0$$

$$\Rightarrow 3\cos^4 \theta - 3\cos^2 \theta + 2\sin^2 \theta - 2\sin^6 \theta = 0$$

$$\Rightarrow 3\cos^2 \theta (\cos^2 \theta - 1) + 2\sin^2 \theta (\sin^4 \theta - 1) = 0$$

$$\Rightarrow -3\cos^2 \theta \sin^2 \theta + 2\sin^2 \theta (1 + \sin^2 \theta) \cos^2 \theta - 1$$

$$\Rightarrow \sin^2 \theta \cos^2 \theta (2 + 2\sin^2 \theta - 3) = 0$$

$$\Rightarrow \sin^2 \theta \cos^2 \theta (2\sin^2 \theta - 1) = 0$$

$$(C1) \sin^2 \theta = 0 \rightarrow 3 \text{ solution; } \theta = \{0, \pi, 2\pi\}$$

$$(C2) \cos^2 \theta = 0 \rightarrow 2 \text{ solution; } \theta = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$$

$$(C3) \sin^2 \theta = \frac{1}{2} \rightarrow 4 \text{ solution; } \theta = \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$

No. of solution = 9

12. Let α and β be two real roots of the equation $(k+1) \tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, where $k (\neq -1)$ and λ are real numbers. If $\tan^2 (\alpha + \beta) = 50$, then a value of λ is :

माना समीकरण $(k+1) \tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, $k (\neq -1)$, $\lambda \in \mathbb{R}$ के α तथा β दो वास्तविक मूल हैं। यदि $\tan^2 (\alpha + \beta) = 50$ है, तो λ का एक मान है :

- (A) $5\sqrt{2}$ (B) $10\sqrt{2}$
(C) 10 (D) 5

Ans. C

Sol. $\tan \alpha + \tan \beta = \frac{\sqrt{2} \lambda}{k+1}$

and $\tan \alpha \tan \beta = \frac{k-1}{k+1}$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{\frac{\sqrt{2} \lambda}{k+1}}{1 - \frac{k-1}{k+1}} = \frac{\lambda}{\sqrt{2}}$$

$$\tan^2 (\alpha + \beta) = 50 = \frac{\lambda^2}{2}$$

$$\Rightarrow \lambda^2 = 100 \Rightarrow \lambda = \pm 10.$$

13. $96 \cos \frac{\pi}{33} \cos \frac{2\pi}{33} \cos \frac{4\pi}{33} \cos \frac{8\pi}{33} \cos \frac{16\pi}{33}$ is equal to :

$96 \cos \frac{\pi}{33} \cos \frac{2\pi}{33} \cos \frac{4\pi}{33} \cos \frac{8\pi}{33} \cos \frac{16\pi}{33}$ बराबर है :

- (A) 4 (B) 1 (C) 2 (D) 3

Ans. D

Sol. Let $\frac{\pi}{33} = A$

$$96 \cos A \cos 2A \cos 2^2 A \cos 2^3 A \cos 2^4 A$$

$$= 96 \frac{\sin 2^5 A}{2^5 \sin A} = \frac{96 \cdot \sin \frac{32\pi}{33}}{32 \sin \frac{\pi}{33}} = \frac{96 \cdot \sin \left(\pi - \frac{\pi}{33} \right)}{32 \sin \frac{\pi}{33}} = \frac{96 \cdot \sin \frac{\pi}{33}}{32 \sin \frac{\pi}{33}} = 3 \text{ Ans.}$$

14. If the solution of the equation $\log_{\cos x} \cot x + 4 \log_{\sin x} \tan x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1} \left(\frac{\alpha + \sqrt{\beta}}{2} \right)$ where α, β are integers, then $\alpha + \beta$ is equal to :

यदि समीकरण $\log_{\cos x} \cot x + 4 \log_{\sin x} \tan x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$ का हल $\sin^{-1} \left(\frac{\alpha + \sqrt{\beta}}{2} \right)$ हैं, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है :

- (A) 6 (B) 3 (C) 4 (D) 5

Ans. C

Sol. $\log_{\cos x} \cot x + 4 \log_{\sin x} \tan x = 1$

$$\frac{\log \cos x - \log \sin x}{\log \cos x} + 4 \frac{(\log \sin x - \log \cos x)}{\log \sin x} = 1$$

$$(\log \sin x)^2 - 4(\log \sin x)(\log \cos x) + 4(\log \cos x)^2 = 1$$

$$\log \sin x = 2 \log \cos x$$

$$\sin^2 x + \sin x - 1 = 0$$

$$\sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\alpha = -1, \beta = 5$$

$$\alpha + \beta = 4$$

15. If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x+y) = \frac{3}{2}$, then $\sin x + \cos y$ is equal to :

यदि $0 < x, y < \pi$ तथा $\cos x + \cos y - \cos(x+y) = \frac{3}{2}$ हैं, तो $\sin x + \cos y$ बराबर है :

- (A) $\frac{1+\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1-\sqrt{3}}{2}$ (D) $\frac{1}{2}$

Ans. A

Sol. $\cos x + \cos y - \cos(x+y) = \frac{3}{2}$

$$2 \cos \frac{x+y}{2} \cos \frac{x-y}{2} - 2 \cos^2 \frac{x+y}{2} + 1 = \frac{3}{2} \quad \dots\dots\dots(1)$$

$$2 \cos \frac{x+5}{2} \cdot 2 \sin \frac{x}{2} \sin \frac{y}{2} = \frac{1}{2} \quad \left\{ \begin{array}{l} 0 < x < \pi \\ 0 < y < \pi \end{array} \right\}$$

$$\left. \begin{array}{l} \sin \frac{x}{2} \rightarrow \text{positive} \\ \sin \frac{y}{2} \rightarrow \text{positive} \end{array} \right\} \text{so } \cos \frac{x+y}{2} \text{ is also positive}$$

$$1 \geq \cos \frac{x-y}{2}$$

$$2 \cos \frac{x+y}{2} \geq 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2} \quad \left(\because \cos \frac{x+y}{2} > 0 \right)$$

$$2 \cos \frac{x+y}{2} - 2 \cos^2 \left(\frac{x+y}{2} \right) + 1 \geq 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2} - 2 \cos^2 \left(\frac{x+y}{2} \right) + 1$$

$$2 \cos \frac{x+y}{2} - 2 \cos^2 \left(\frac{x+y}{2} \right) + 1 \geq \frac{3}{2}$$

$$\left(2 \cos \left(\frac{x+y}{2} \right) - 1 \right)^2 \leq 0$$

$$\cos \left(\frac{x+y}{2} \right) = \frac{1}{2}$$

$$\frac{x+y}{2} = \frac{\pi}{3}$$

$$x+y = \frac{2\pi}{3}$$

from equation (1)

$$2 \cos \frac{\pi}{3} \cos \frac{x-y}{2} - 2 \cos^2 \frac{\pi}{3} + 1 = \frac{3}{2}$$

$$\cos \frac{x-y}{2} = 1$$

$$x-y = 0$$

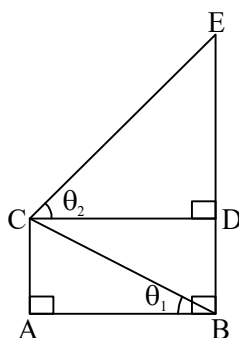
$$x = y$$

put $x = y = \frac{\pi}{3}$

INTEGER TYPE QUESTIONS

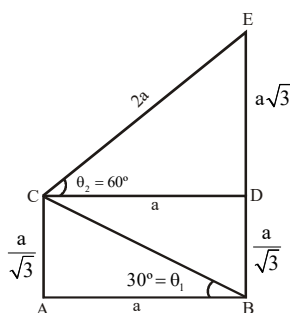
16. In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(BE) = 4(AB)$. If the area of ΔCAB is $2\sqrt{3} - 3$ unit², when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of ΔCED is equal to _____.

दी गई आकृति में $\theta_1 + \theta_2 = \frac{\pi}{2}$ तथा $\sqrt{3}(BE) = 4(AB)$ है। यदि ΔCAB का क्षेत्रफल $2\sqrt{3} - 3$ वर्ग इकाई है, जब $\frac{\theta_2}{\theta_1}$ अधिकतम है, तो ΔCED का परिमाप (इकाई में) बराबर है :



Ans. 6

Sol.



$$\tan \theta_2 = \frac{DE}{CD} = \frac{BE - AC}{AB}$$

$$\tan \theta_2 = \frac{4}{\sqrt{3}} - \tan \theta_1 \Rightarrow \tan \theta_1 + \tan \theta_2 = \frac{4}{\sqrt{3}} \dots\dots$$

$$\text{Now } \theta_1 + \theta_2 = \frac{\pi}{2} \Rightarrow \tan \theta_1 \tan \theta_2 = 1 \dots (2)$$

Let $\tan \theta_1 = x$ and $\tan \theta_2 = y$ and $\frac{\theta_2}{\theta_1}$ is the largest

$$\Rightarrow x + y = \frac{4}{\sqrt{3}} \quad x + \frac{1}{x} = \frac{4}{\sqrt{3}} \quad \sqrt{3}x^2 - 4x + \sqrt{3} = 0$$

$$xy = 1$$

$$\sqrt{3}x(x - \sqrt{3}) - 1(x - \sqrt{3}) = 0$$

$$x = \sqrt{3}, x = \frac{1}{\sqrt{3}}$$

$$y = \frac{1}{\sqrt{3}}, y = \sqrt{3}$$

$$x = 60^\circ, y = 30^\circ$$

$$\frac{x}{y} = 2$$

17. Let $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos \theta) + \tan(\pi \sin \theta) = 0\}$. Then $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to _____.

माना $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos \theta) + \tan(\pi \sin \theta) = 0\}$ है। तब $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ बराबर है _____

Ans. 2

Sol. $\tan(\pi \cos \theta) + \tan(\pi \sin \theta) = 0$

$$\tan(\pi \cos \theta) = -\tan(\pi \sin \theta)$$

$$\tan(\pi \cos \theta) = \tan(-\pi \sin \theta)$$

$$\pi \cos \theta = n\pi - \pi \sin \theta, n \in \mathbb{I}$$

$$\sin \theta + \cos \theta = n$$

$$\therefore \sin \theta + \cos \theta \in [-\sqrt{2}, \sqrt{2}]$$

Then possible value of $n = -1, 0, 1$

$$\text{Now } \sin \theta + \cos \theta = 1$$

$$\theta = 0, \frac{\pi}{2},$$

$$\begin{cases} \sin \theta + \cos \theta = 0 \\ \tan \theta = 1 \\ \theta = \frac{3\pi}{4}, \frac{7\pi}{4}, \end{cases}$$

$$\sin \theta + \cos \theta = -1$$

$$\theta = \frac{3\pi}{2}, \pi$$

$$\text{So, } \sum_{\theta \in S} \sin^2 \left(\theta + \frac{\pi}{4} \right) = 4 \left(\frac{1}{2} \right) + 2(0) = 2$$

18. If $\frac{\sqrt{2} \sin \alpha}{\sqrt{1 + \cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1 - \cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$, $\alpha, \beta \in \left(0, \frac{\pi}{2} \right)$, then $\tan (\alpha + 2\beta)$ is equal to _____.

यदि $\frac{\sqrt{2} \sin \alpha}{\sqrt{1 + \cos 2\alpha}} = \frac{1}{7}$ तथा $\sqrt{\frac{1 - \cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$, $\alpha, \beta \in \left(0, \frac{\pi}{2} \right)$ हैं, तो $(\alpha + 2\beta)$ बराबर है _____ ।

Ans. 1

Sol. $\frac{\sqrt{2} \sin \alpha}{\sqrt{1 + \cos 2\alpha}} = \frac{1}{7}$ $\alpha, \beta \in \left(0, \frac{\pi}{2} \right)$

$$\frac{\sqrt{2} \sin \alpha}{\sqrt{2 \cos \alpha}} = \frac{1}{7} \Rightarrow \tan \alpha = \frac{1}{7}$$

$$\sqrt{\frac{1 - \cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$$

$$\sin \beta = \frac{1}{\sqrt{10}}$$

$$\tan \beta = \frac{1}{3}$$

$$\tan (\alpha + \beta) = \frac{\tan \alpha + \tan 2\beta}{1 - \tan \alpha \cdot \tan 2\beta}$$

$$\tan 2\beta = \frac{2 \cdot \frac{1}{3}}{1 - \frac{1}{9}} = \frac{3}{4}$$

$$\tan (\alpha + 2\beta) = \frac{\frac{1}{7} + \frac{3}{4}}{1 - \frac{1}{7} \times \frac{3}{4}}$$

$$\tan(\alpha + 2\beta) = \frac{25}{25} = 1$$